1. Team Identity

1.1 Studio / Group Name

[The Lab Rats]

1.2 Team Members

Name	Primary Role	Secondary Role	Strengths Brought to the Project
[Nandita Uniyal]	[Coding / Wiring/ Fabrication / Mechanics/]	[aesthetics , 3d printing files]	[Strong creative direction, attention to detail, quick learning, ability to translate ideas into physical form]
[Shania Singh]	[Electronics / aesthetics/ Fabrication/ Mechanics]	[Coding / laser cutting files]	System thinking, problem-solving, hands-on execution, ability to integrate technical and physical components

1.3 Project Title

[Colour Chaos Lab]

1.4 One-Line Pitch

[A simulated laboratory setup where two chemical solutions are combined in a conical flask, triggering a reaction that reveals an unexpected transformation to demonstrate cause, effect, and perceptual surprise.]

1.5 Expanded Project Idea

In 1–2 paragraphs, explain:

- what your project is,
- what kind of playful experience it creates,
- what makes it fun, curious, engaging, strange, satisfying, competitive, or delightful,
- what technologies are involved.

Response:

`[1. What the project is A simulated laboratory installation where users mix different chemical solutions in a controlled setup, observing real-time reactions that demonstrate transformation, unpredictability, and cause-and-effect within a safe, fabricated environment.

2. The playful experience The experience gives users complete freedom to experiment, encouraging curiosity and trial-and-error as they combine substances without fixed outcomes, creating a sense of childlike exploration, suspense, and anticipation around what might happen next.

3. Technology involved An interactive app interface enables users to control the experiment through sliders, adjusting the quantity and flow of each chemical via connected servo mechanisms that regulate the pouring process into the conical flask.]`

2. Philosophy Fit

2.1 Experience, Not Social Problem

This module does **not** require your project to solve a large social problem.

You are allowed to build:

- toys,
- games,
- interactive objects,

- playful machines,
- kinetic artifacts,
- humorous devices,
- strange but delightful experiences,
- things that are entertaining to use or watch.

2.2 What kind of experience are you creating?

Answer the following:

- What is the experience?
- What do you want the player or participant to feel?
- Why would someone want to try it again?

Response:

` [What is the experience? An interactive, self-directed experiment where participants actively combine different chemical inputs and observe immediate reactions, turning the process into a continuous loop of testing, observing, and adjusting outcomes.

What do you want the player or participant to feel? Intrigued, surprised, and excited.

Why would someone want to try it again? To explore alternative combinations, uncover a wider range of outcomes, and gradually develop control over the results, motivating repeated interaction through experimentation and refinement.]`

2.3 Design Persona

Complete the sentence below:

We are designing this project as if we are a small creative studio making a **[toy / game / playable object / interactive experience]** for **[children / teens / adults / classmates / exhibition visitors / mixed audience]**.

Response:

[We are designing this project as if we are a small creative studio making a interactive experience for teens and exhibition visitors .]

3. Inspiration

3.1 References

List what inspired the project.

Source Type	Title / Link	What Inspired You
[Toy]	[Chemistry experiment kits]	[Open-ended experimentation, mixing-based interaction, and visible cause-and-effect outcomes]
[Object]	[Interactive installations]	[Combining physical movement with responsive feedback to create immersive, engaging experiences]
[references]	[https://pin.it/5HaTeSwcE , https://pin.it/6rlKVXC76]	[Visual composition, layout of elements, and creating a detailed, believable lab-like environment]

3.2 Original Twist

What makes your project original?

Response:

[It’s original because it doesn’t follow a fixed outcome or scripted interaction—you’re directly controlling the experiment and discovering results in real time. The combination of a physical setup with digital control, and the fact that every outcome depends on your input, makes the experience inherently unpredictable and uniquely yours every time.]

4. Project Intent

4.1 Core Interaction Loop

Describe the main loop of interaction.

Examples:

- press → launch → score → reset
- connect → control → observe → repeat
- turn → trigger → react → repeat
- move object → sensor detects → sound/light response → player reacts

Response:

[control → dispense → react → observe → repeat]

4.2 Intended Player / Audience

Question	Response
Who is this for?	[Curious minds]
Age range	[12 – 20 yrs]
Solo or multiplayer	[Solo]
Expected duration of one round	[1–2 minutes]
What should the player feel?	[Intrigued, surprised, and excited]

Is explanation required before use?	[Minimal intrustions]

4.3 Player Journey

Describe exactly how a player will use the project.

1. **Approach:** [The player notices a lab setup with a conical flask and an interface, where sensor-activated doors open as they come closer, inviting interaction.]
2. **Start:** [They access the control interface, preparing to begin the experiment.]
3. **First Action:** [They select and adjust the quantities of the two chemicals using sliders.]
4. **Main Interaction:** [They continuously modify inputs, testing different combinations and observing how each change affects the outcome.]
5. **System Response:** [The system dispenses the chemicals into the flask, triggers a visible reaction, and responds with flashing lights at the moment of reaction, revealing an unexpected result in real time.]
6. **Win / Lose / End Condition:** [A round ends once the reaction is complete and the final outcome is revealed, with no fixed win or lose state.]
7. **Reset:** [The flasks are manually reset for the next round, while the lighting system automatically resets, preparing the setup for a new experiment.]

4.4 Rules of Play

If your project is a game, list the rules clearly.

Not applicable. The project is an interactive experience, not a game. The interaction follows basic factual constraints, such as users controlling the experiment only through the provided interface and not directly interacting with the materials.

5. Definition of Success

5.1 Definition of “Playable”

Your project will be considered complete only if these conditions are met.

- ` [The user can independently start and control the experiment through the interface

]`

- [Chemical dispensing responds accurately to user input (sliders/controls)]
- [A visible reaction is produced consistently after each interaction]
- [Sensor-based elements (doors, lights) function reliably during the experience]
- [The system can be reset (manual flask reset + automatic light reset) for repeated use]

5.2 Minimum Viable Version

What is the smallest version of this project that still delivers the core experience?

Response:

[A basic setup with one conical flask and two controlled liquid inputs, where users adjust simple sliders to mix them and observe a visible reaction, without additional effects but retaining core interaction and outcome.]

5.3 Stretch Features

What features are nice to have but not essential?

- [Enhanced aesthetics and detailed miniature elements]
- [LEDs changing color according to the liquid]
- [patterns of lighting]
- [The writing of “chemical chaos” on the set up]

6. System Overview

6.1 Project Type

Check all that apply.

- [YES] Electronics-based
- [YES] Mechanical
- [YES] Sensor-based
- [NO] App-connected
- [YES] Motorized
- [NO] Sound-based
- [YES] Light-based
- [NO] Screen/UI-based
- [YES] Fabricated structure
- [NO] Game logic based
- [YES] Installation / tabletop experience
- Other: [Write here]

6.2 High-Level System Description

Explain how the system works in simple terms.

Include:

- input,
- processing,
- output,
- physical structure,
- app interaction if any.

Response:

[Input is given through buttons, which send signals to a microcontroller. It processes these signals to control servo motors that rotate the test tubes and dispense liquids into a flask. The physical setup includes the base, tubes, flask, sensors, and LEDs. The output is a

visible reaction, with LEDs turning pink and the scientist figure moving. There is no app interaction.]

6.3 Input / Output Map

System Part	Type	What It Does
[Button]	Input	[Sends signal when pressed to control servo movement]
[ESP32]	Processing	[Reads inputs and controls servos and LED response]
[LED]	Output	[Lights up (turns pink) to indicate reaction completion]
[Servo]	Physical Action	[Rotates test tubes in steps to dispense liquids]
[power supply]	Support	[Provides stable power to all components]

7. Sketches and Visual Planning

7.1 Concept Sketch

Add an early sketch of the full idea.

Insert image below:

[Upload image and link here]

Example:

7.2 Labeled Build Sketch

Add a sketch with labels showing:

- structure,
- electronics placement,
- user touch points,
- moving parts,
- output elements.

Insert image below:

[Upload image and link here]

7.3 Approximate Dimensions

Dimension	Value
Length	[28cm]
Width	[48cm]
Height	[28cm]

Estimated weight	[750g]

8. Mechanical Planning

8.1 Mechanical Features

Check all that apply.

- Gears
- Pulleys
- Belt drives
- Linkages
- Hinges
- Shafts
- Springs
- Bearings
- Wheels
- Sliders
- Levers
- [YES.] Not applicable

8.2 Mechanical Description

Describe the mechanism and what it is meant to do.

Response: [The mechanism uses two servo motors attached to test tubes that rotate towards each other to dispense liquids into a conical flask. Each servo is controlled by a button, and every press rotates the test tube by 10 degrees, allowing controlled pouring.

The motion continues in steps until both test tubes reach up to 100 degrees, enabling the liquids to mix and produce the final reaction, which triggers the scientist figure to wake up and causes the lights to flash.]

8.3 Motion Planning

If something moves, explain:

- what moves,
- what causes the movement,
- how far it moves,
- how fast it moves,
- what could go wrong.

Response:

`[

- what moves- Both test tubes rotate towards each other, each moving up to 120 degrees using the buttons
- what causes the movement- Pressing the buttons sends signals to the servo motors, triggering the motion
- how far it moves- Each test tube rotates up to 120 degrees, one moving left and the other right
- how fast it moves- The movement happens in steps of 30 degrees with each button press, the angle it reset to original place by pressing boththe buttons.
- what could go wrong- With prolonged use, the servo motors may get stuck or lose precision

8.4 Simulation / CAD / Animation Before Making

If your project includes mechanical motion, document the digital planning before fabrication.

NA

8.5 Changes After Digital Testing

What changed after the CAD, animation, or simulation stage?

Response:

'[We learned the limitations and permanence of materials, especially after processes like laser cutting and 3D printing, where changes are not reversible.We had to adapt the design based on how parts actually fit together in reality, not just in concept. EX- we changed the base structure from a simple square to a more stable form that fit in all components. EX- shifted the servo placement from the corners to a flat surface for better alignment and positioning.]'

9. Electronics Planning

9.1 Electronics Used

Component	Quantity	Purpose
[ESP32]	1	[Main controller]
[Breadboard power supply]	[1]	[additional power]
[servo motor]	[3]	[left right movement]
[led strip]	[70 PIXELS(1)]	[enhancment and lighting up]
[push button]	[2]	[control servos]

9.2 Wiring Plan

Describe the main electrical connections.

Response:

[Microcontroller: ESP32 acts as the central controller.

Servos:

Servo1 signal → GPIO 18

Servo2 signal → GPIO 19

Servo3 signal → GPIO 23

All servos share common GND

Powered using external 5V supply

Buttons:

Button1 → GPIO 4

Button2 → GPIO 5

Other terminal of buttons → GND

NeoPixel LED Strip:

Data pin → GPIO 21

VCC → 5V external supply

GND → common ground

Grounding:

ESP32 GND, power supply GND, servos, and LED strip are all connected together

]

9.3 Circuit Diagram

Insert a hand-drawn or software-made circuit diagram.

Insert image below:

[Upload image and link here]

9.4 Power Plan

Question	Response

Power source	[External 5V adapter + USB (ESP32)]
Voltage required	[5V for servos and LED strip]
Current concerns	[Servos and LED strip draw high current; ESP32 cannot supply enough power alone thats why we use a bread board power supply]
Safety concerns	[Avoid powering servos from ESP32 directly; ensure common ground; prevent overheating and short circuits]

10. Software Planning

10.1 Software Tools

Tool / Platform	Purpose
[MicroPython]	[Writing and running code on ESP32]
[Thonny IDE, Servo Library] NeoPixel Library}	[Thonny IDE: Uploading and debugging code Servo Library: Controlling servo motor angles Neo Pixel Library: Control

	ling LED strip animations]

10.2 Software Logic

Describe what the code must do.

Include:

- startup behavior,
- input handling,
- sensor reading,
- decision logic,
- output behavior,
- communication logic,
- reset behavior.

Response:

[Startup Behavior:

Initialize all pins (buttons, servos, LED strip)

Set initial servo positions:

Servo1 → 0°

Servo2 → 180°

Servo3 → 0°

Turn off all LEDs

Input Handling:

Continuously read button states

Detect button press using edge detection (HIGH → LOW transition)

Sensor Reading:

No external sensors used in final system

Only digital input from buttons

Decision Logic:

If Button1 pressed → increase Servo1 angle (max 100°)

If Button2 pressed → decrease Servo2 angle (max 70°)

If both buttons pressed → reset system

If Servo1 = 100° AND Servo2 = 70° → trigger final sequence

Output Behavior:

Servo1 and Servo2 move step-wise based on button input

LED strip performs pink chase animation

LED strip turns OFF

Servo3 performs sweeping motion (0° → 180°)

Communication Logic:

Serial print statements used for debugging (angle values, reset, trigger events)

Reset Behavior:

When both buttons are pressed:

Reset servo angles to initial positions

Turn off LEDs

Allow system to restart interaction]

10.3 Code Flowchart

Insert a flowchart showing your code logic.

Suggested sequence:

- start,
- initialize,
- wait for input,
- read input,
- decision,
- trigger output,
- repeat or reset,
- error handling.

Insert image below:

[Upload image and link here]

10.4 Pseudocode

[start program

setup servos, buttons, LEDs

set servo1 = 0, servo2 = 180, servo3 = 0

turn LEDs off

```
while True:

# read buttons

# if both buttons pressed
   # reset all servos to start
   # turn LEDs off

# if button1 pressed
   # increase servo1 (max 100)
```

```
# if button2 pressed
   # decrease servo2 (min 70)

# move servo1 and servo2

# if servo1 = 100 and servo2 = 70
   # run LED animation (on then off)
   # move servo3 from 0 to 170
   # do this only once

# small delay
]
```

11. MIT App Inventor Plan

11.1 Is an app part of this project?

- Yes
- No

If yes, complete this section.

NOT APPLICABLE

12. Bill of Materials

12.1 Full BOM

Item	Quantity	In Kit?	Need to Buy?	Estimated Cost	Why This Choice?
[ESP32]	1	Yes	[YES]	[400]	[ESP in kit was weak and old]

[Power supply]	[1]	[Yes]	[YES]	[100]	[Power supply is kit was fried and weak]
[Servo]	[4]	[No]	[YES]	[600]	[need more amounts and ones in the kit were weak]
[LED strip]	[1]	[No]	[YES]	[200]	[To enhance the experience also reveal the colour of the solution]
[MINI breadboard]	[1]	[No]	[YES]	[50]	[for putting the buttons-making the controller]

12.2 Material Justification

Explain why you selected your main materials and components.

Examples:

- Why acrylic instead of cardboard?
- Why MDF instead of 3D print?
- Why servo instead of DC motor?
- Why bearing instead of a plain shaft hole?

Response:

[Why acrylic instead of cardboard?- Acrylic was chosen for its transparency, allowing clear visibility of the internal setup, and for its higher stiffness and durability compared to cardboard, making the structure more stable.

Why MDF instead of 3D print?-

MDF was more suitable for the structural components as it provides better strength and stability. It was also easier to fabricate and assemble

Why servo instead of DC motor?- Servo motors were used because they allow precise, controlled movement at fixed angles, which is necessary for accurate dispensing.
DC motor is also heavy and the servo motors complete the task more effectively.

12.3 Items to Purchase Separately

Item	Why Needed	Purchase Link	Latest Safe Date to Procure	Status
[chemicals]	[for the experiment]	[from shop-in ulhasnagar]	[10/4/26]	[Received]
[beakers]	[pouring liquid in the experiment]	[same shop]	[10/4/26]	[Received]
[clear bottles]	[interior decoration]	[stationary shop]	[15/4/26]	[Received]
[decorative miniatures]	[interior decoration and aesthetics]	[shop]	[16/4/26]	[Received]

12.4 Budget Summary

Budget Item	Estimated Cost
Electronics	[1400]

other parts	[700]
Purchased extras	[800]
Total	[2900]

12.5 Budget Reflection

If your cost is too high, what can be simplified, removed, substituted, or shared?

Response:

[It feels accurate, especially due to the strong focus on aesthetics and detailing. Additional costs came from purchasing technical components that were not provided by the college, replacing older or weaker parts ourselves.]

13. Planning the Work

13.1 Team Working Agreement

Write how your team will work together.

Include:

- how tasks are divided,
- how decisions are made,
- how progress will be checked,
- what happens if a task is delayed,
- how documentation will be maintained.

Response:

[Tasks are divided based on strengths, with both members taking responsibility for execution and support. Decisions are made through quick discussion and agreement. Progress is checked regularly during work sessions. If a task is delayed, the other member steps in to assist and adjust timelines. Documentation is maintained alongside the process through notes, photos, and recorded updates.]

13.2 Task Breakdown

Task ID	Task	Owner	Estimated Hours	Deadline	Status
T1	[Finalize concept]	[BOTH]	2	[2nd week]	DONE
T2	[Complete BOM]	[Shania]	15ins	[19/4/26]	DONE
T3	[Test electronics]	[Nandita]	6-8	[12/4/26]	DONE
T4	[Build structure]	[BOTH]	24	[16/4/26]	DONE
T5	[Write control code]	[Nandita]	6	[10/4/26]	DONE
T6	[Integrate system]	[Shania]	4	[10/4/26]	DONE
T7	[Playtest]	[BOTH]	2	[16/4/26]	DONE
T8	[Refine and document]	[BOTH]	2	[17/4/26]	DONE

13.3 Responsibility Split

Area	Main Owner	Support Owner
Concept and gameplay	[BOTH]	[BOTH.]
Electronics	[SHANIA]	[NANDITA]
Coding	[NANDITA]	[SHANIA]
Mechanical build	[NANDITA]	[SHANIA]
Testing	[SHANIA]	[NANDITA]

Documentation	[SHANIA]	[NANDITA]

14. Weekly Milestones

14.1 Four-Week Plan

Week 1 — Plan and De-risk

Expected outcomes:

- Idea finalized
- Core interaction decided
- Sketches made
- BOM completed
- Purchase needs identified
- Key uncertainty identified
- Basic feasibility tested

Week 2 — Build Subsystems

Expected outcomes:

- Electronics tests completed
- CAD / structure planning completed
- App UI started if needed
- Mechanical concept tested
- Main subsystems partially working

Week 3 — Integrate

Expected outcomes:

- Physical body built
- Electronics integrated
- Code connected to hardware

- App connected if required
- First playable version exists

Week 4 — Refine and Finish

Expected outcomes:

- Technical bugs reduced
- Playtesting completed
- Improvements made
- Documentation completed
- Final build ready

14.2 Weekly Update Log

Week	Planned Goal	What Actually Happened	What Changed	Next Steps
Week 1	[Finalize concept & basic structure]	[Concept discussed & roughly finalized; initial sketches and layout started]	[Scope slightly reduced to keep it achievable in time]	[Begin prototyping, test basic mechanisms]

| Week 2 | [Working on the code, testing and buying parts and materials] | [ordering parts, starting cad files] | [Design of base and mounting was modified for better support] | [Refine structure and improve component placement] |

| Week 3 | [Integrate components and code+work on aesthetics] | [Components were added but required adjustments and reworking] | [Changed positioning of parts for better functionality and alignment] | [Finalize working model and fix inconsistencies] |

| Week 4 | [Final assembly and testing] | [Model was completed with some last-minute fixes, turned out better than expected.] | [Small placement adjustments were made during final stage] | [Finish up repo details] |

15. Risks and Unknowns

15.1 Risk Register

Risk	Type	Likelihood	Impact	Mitigation Plan	Owner
[Servo motors getting stuck or misaligned]	Technical	Medium	High	[Recalibrate angles, secure mounting, replace faulty servos]	[Name]
[Structure instability or misfit parts]	material	high	medium	[Reinforce base, adjust joints, rework dimensions]	[Name]
[LED not syncing with reaction]	[Time]	[Medium]	[Low]	[Debug code timing and check wiring connections]	[Name]
[Button input not responding consistently]	[Technical]	[High]	[Low/Medium/High]	[Check connections, replace faulty buttons, ensure proper wiring]	[Name]

15.2 Biggest Unknown Right Now

What is the single biggest uncertainty in your project at this stage?

Response:

[Consistency of the reaction output and synchronization between servo movement and LED response.]

16. Testing and Playtesting

16.1 Technical Testing Plan

What Needs Testing	How You Will Test It	Success Condition
[Servo 1 and 2 (movement of test tubes)]	[Press buttons repeatedly and observe motion]	[Smooth rotation in fixed steps without delay or misalignment]
[servo 3 (movement of scientist)]	[Trigger after reaction and observe motion]	[Scientist moves at the correct moment without lag]
[Timing and colour of LED strip]	[Run full experiment and observe LED response]	[LEDs turn pink exactly at reaction completion]
[Connection of the buttons with the servos]	[Press buttons multiple times and see reactions]	[Immediate and consistent response every time]

16.2 Playtesting Plan

Question	How You Will Check
Do players understand what to do?	[we will use a separate instruction manual and check that it is understandable/simple]
Is the interaction satisfying?	[by the reaction of the players and the accuracy of the experiment, Ask for feedback after one full cycle]
Do players want another turn?	[See if they repeat interaction voluntarily]
Is the challenge balanced?	[Check if users can control outcome without confusion]
Is the response clear and immediate?	[Observe reaction timing and user response]

16.3 Testing and Debugging Log

Date	Problem Found	Type	What You Tried	Result	Next Action
[Date]	[Servo jitter during movement]	[Mechanical]	[Adjusted code delay/ replaced old weak servos]	[Partly]	[Further calibration]

[Date]	[LED delay after reaction]	[Technical]	[Modified timing in code]	[Worked]	[Fine-tune response]
	[base not fitting all items		[reforming the shape of the base to adjust all parts]	[Worked]	Stabize the additions

16.4 Playtesting Notes

Tester	What They Did	What Confused Them	What They Enjoyed	What You Will Change
[friend]	[Interacted with buttons and completed experiment]	[Slight confusion on how much/long to press]	[Reaction of chemicals and movement f the figure]	[Add clearer instruction]
[classmate]	[Tried multiple combinations]	[Delay in understanding control]	[Repetition and surprise element]	[Improve responsivenes]

17. Build Documentation

17.1 Fabrication Process

Describe how the project was physically made.

Include:

- cutting,
- 3D printing,
- assembly,
- fastening,
- wiring,
- finishing,
- revisions.

Response:

[For this project, the first step was preparing the DXF files for laser cutting MDF and 3D printing. This included taking measurements and deciding the placement of all technical elements. After that was completed, we arranged the base and created the basic structure. We then made the miniature elements (bookshelf, two tables, and a round table for the scientist), followed by all other decorative elements. We tested all the technical parts separately, and then combined them by integrating the code. After that, the wiring was completed and the final pins were decided and noted down. All elements were assembled and testing began. We tested the movement of the servos, the working of the buttons, different LED light combinations, and the physical working of the chemical experiment. Finally, all decisions were photographed and recorded for proof and future reference.]

17.2 Build Photos

Add photos throughout the project.

Suggested images:

- early sketch,
- prototype,
- electronics testing,
- mechanism test,
- app screenshot,
- final build.

17.3 Version History

Version	Date	What Changed	Why
v1	[Initial stage]	Basic square base, initial servo placement, simple lighting concept	Early prototype to test feasibility and core mechanism
v2	Mid development	Changed base to hexagonal form, improved servo positioning, integrated button control	To improve stability, alignment, and usability
v3	[Final]	Refined structure, synchronized LED response (pink on reaction), added scientist movement	[To enhance clarity of output and overall experience]

18. Final Outcome

18.1 Final Description

Describe the final version of your project.

Response:

[17.3 Version History

Version	Date	What Changed	Why
v1	Initial stage	Basic square base, initial servo placement, simple lighting concept	Early prototype to test feasibility and core mechanism
v2	Mid development	Changed base to hexagonal form, improved servo positioning, integrated button control	To improve stability, alignment, and usability
v3	Final	Refined structure, synchronized LED response (pink on reaction), added scientist movement	To enhance clarity of output and overall experience

18. Final Outcome

18.1 Final Description

Response: The final project is a fully functional interactive installation simulating a laboratory experiment. Users engage with the system through physical buttons, which control servo-driven test tubes that dispense liquids into a central conical flask. The setup is housed within a structured enclosure designed with detailed miniature elements, creating a cohesive lab-like environment.

The system integrates mechanical movement, electronics, and visual feedback to produce a clear and engaging outcome. As the liquids combine, a visible reaction occurs, accompanied by LED lighting that turns pink to indicate completion. Simultaneously, a scientist figure is activated, adding a narrative and performative layer to the interaction.

The project successfully combines physical interaction, controlled mechanisms, and responsive outputs to create an intuitive and immersive experience.

]

18.2 What Works Well

- [LED lights responding and changing color during the pouring process]
- [Clear and reliable connection between the buttons and both servo motors]
- [Refined and synchronized movement of the scientist figure]
- [Consistent execution of the experiment and visible color transformation of the solutions]

18.3 What Still Needs Improvement

- [**Placement of the tubes** – Better alignment and positioning to ensure smoother and more accurate pouring]
- [**Coordination of the servo motors** – Improved synchronization to make the movement more consistent and controlled]
- [**Adding more technical elements** – Expanding the system with additional sensors or feedback to enhance interaction and responsiveness]

18.4 What Changed From the Original Plan

How did the project change from the initial idea?

Response:

[The project evolved significantly from the initial idea as we began testing and building. The base was originally a simple square, but we changed it to a hexagonal structure to better accommodate the components and improve overall stability and fitting.

Initially, lighting was only considered as a general ambient feature to enhance the environment. During development, this was refined into a more responsive element, where the lights flash pink specifically at the moment the reaction is completed, making the outcome more clear and impactful.

We also adjusted the placement of components like servos and test tubes to improve alignment and functionality, shifting from corner-based mounting to more precise positioning on flat surfaces]

19. Reflection

19.1 Team Reflection

What did your team do well?

What slowed you down?

How well did you manage time, tasks, and responsibilities?

Response:

`[Our team maintained strong coordination and we easily divided tasks based on individual strengths yet still supporting each other throughout our indiviual tasks. This allowed us to sustain a balanced and efficient workflow.

I think one thing that slowed us down was getting distracted as we are quite good friends and enjoy each others company more than work (sometimes...) At times, this reduced overall efficiency during work sessions.

responsibilities and tasks were managed and divided well, We struugled with time constraints, as multiple iterations and reworking of components slowed progress.]`

19.2 Technical Reflection

What did you learn about:

- electronics,
- coding,
- mechanisms,
- fabrication,
- integration?

Response:

`[**Response:**

Electronics Learned how to wire components properly, manage sensors, and maintain consistent system performance

Coding Understood how to translate user input into physical actions and handle real-time control and debugging

Mechanisms Worked on controlling flow and movement using servos, focusing on precision and consistency

Fabrication Learned to build and assemble parts accurately so the system remains stable and functional

Integration Understood how to bring together app control, electronics, and physical components into one seamless working system]`

19.3 Design Reflection

What did you learn about:

- designing for play,
- delight,
- clarity,
- physical interaction,
- player understanding,
- iteration?

Response:

`[Designing for play Play works best when the system is open-ended and not over-instructed. Giving users freedom to explore without fixed outcomes keeps them engaged longer.

Delight Delight comes from timing and surprise. The moment of reaction and the unexpected result create the strongest impact

Clarity The interaction has to be immediately understandable. If users hesitate or feel confused about what to do, the experience breaks

Physical interaction Tangible elements make the experience feel real and immersive. Seeing actual movement, flow, and reaction adds weight compared to a purely digital interaction.

Iteration Small adjustments in timing, flow, and responsiveness make a big difference. Repeated testing helps refine both the technical system and the overall experience.]`

19.4 If You Had One More Week

What would you improve next?

Response:

[We would improve the precision and consistency of the dispensing system to achieve more controlled outcomes. We would add automated doors with sensor-based activation and refine the overall structure for better stability and finish. We would include a wider range of liquids to experiment with along with more varied lighting effects]

20. Final Submission Checklist

Before submission, confirm that:

- [YES] Team details are complete
- [YES] Project description is complete
- [YES] Inspiration sources are included
- [YES]Player journey is written
- [YES] Sketches are added
- [YES] BOM is complete
- [YES] Purchase list is complete
- [YES] Budget summary is complete
- [YES] Mechanical planning is documented if applicable
- [na] App planning is documented if applicable
- [YES] Code flowchart is added
- [YES] Task breakdown is complete
- [YES]Weekly logs are updated
- [YES] Risk register is complete
- [YES] Testing log is updated
- [YES] Playtesting notes are included
- [YES] Build photos are included

- [YES] Final reflection is written

  Thank you please enjoy our project <3

22. Instructor Review

22.1 Proposal Approval

- Approved to proceed
- Approved with changes
- Rework required before proceeding

Instructor comments:

[Instructor fills this section]

22.2 Midpoint Review

[Instructor fills this section]

22.3 Final Review Notes

[Instructor fills this section]